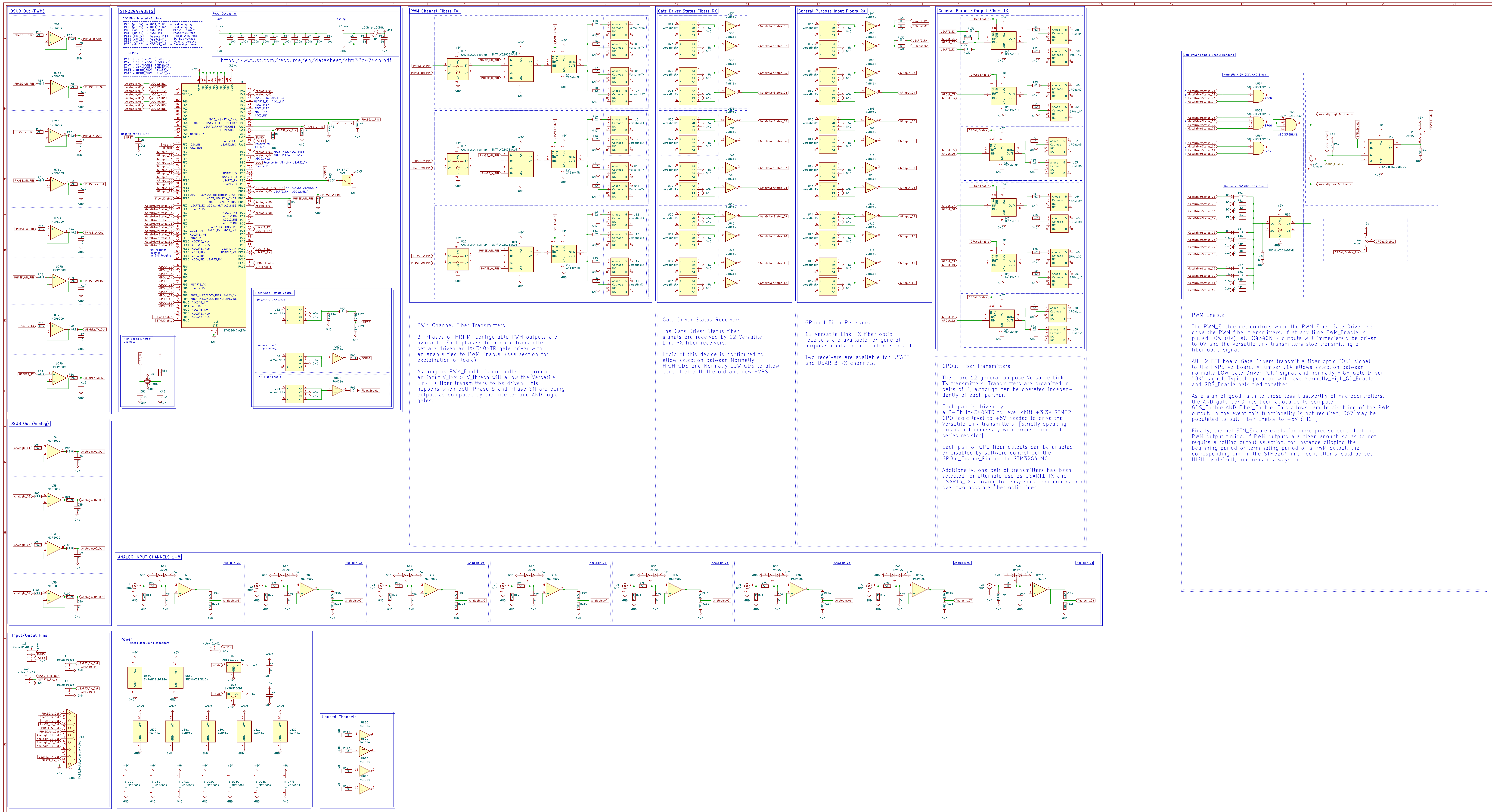
DSUB Out (PWM)
STM32G474QET6
https://www.st.com/resource/en/datasheet/stm32g474cb.pdf
PWM Channel Fibers TX
Gate Driver Status Fibers RX
General Purpose Input Fibers RX
General Purpose Output Fibers TX
Gate Driver Fault & Enable Handling
DSUB Out (Analog)
ANALOG INPUT CHANNELS 1-8
Input/Output Pins
Power
Unused Channels

PWM Channel Fiber Transmitters

3–Phases of HRTIM–configurable PWM outputs are available. Each phase's fiber optic transmitter set are driven an IX4340NTR gate driver with an enable tied to PWM_Enable. (see section for explaination of logic)

As long as PWM_Enable is not pulled to ground an input V_INx > V_thresh will allow the Versatile Link TX fiber transmitters to be driven. This happens when both Phase_S and Phase_SN are being output, as computed by the inverter and AND logic gates.

Gate Driver Status Receivers

The Gate Driver Status fiber signals are received by 12 Versatile Link RX fiber receivers.

Logic of this device is configured to allow selection between Normally HIGH GDS and Normally LOW GDS to allow control of both the old and new HVPS.

GPInput Fiber Receivers

12 Versatile Link RX fiber optic receivers are available for general purpose inputs to the controller board.

Two receivers are available for USART1 and USART3 RX channels.

GPOut Fiber Transmitters

There are 12 general purpose Versatile Link TX transmitters. Transmitters are organized in pairs of 2, although can be operated independently of each partner.

Each pair is driven by a 2–Ch IX4340NTR to level shift +3.3V STM32 GPO logic level to +5V needed to drive the Versatile Link transmitters. [Strictly speaking this is not necessary with proper choice of series resistor].

Each pair of GPO fiber outputs can be enabled or disabled by software control ouf the GPOut_Enable_Pin on the STM32G4 MCU.

Additionally, one pair of transmitters has been selected for alternate use as USART1_TX and USART3_TX allowing for easy serial communication over two possible fiber optic lines.

PWM_Enable:

The PWM_Enable net controls when the PWM Fiber Gate Driver ICs drive the PWM fiber transmitters. If at any time PWM_Enable is pulled LOW (0V), all IX4340NTR outputs will immediately be driven to 0V and the versatile link transmitters stop transmitting a fiber optic signal.

All 12 FET board Gate Drivers transmit a fiber optic "OK" signal to the HVPS V3 board. A jumper J14 allows selection between normally LOW Gate Driver "OK" signal and normally HIGH Gate Driver "OK" signal. Typical operation will have Normally_High_GD_Enable and GDS_Enable nets tied together.

As a sign of good faith to those less trustworthy of microcontrollers, the AND gate U54D has been allocated to compute GDS_Enable AND Fiber_Enable. This allows remote disabling of the PWM output. In the event this functionality is not required, R67 may be populated to pull Fiber_Enable to +5V (HIGH).

Finally, the net STM_Enable exists for more precise control of the PWM output timing. If PWM outputs are clean enough so as to not require a rolling output selection, for instance clipping the beginning period or terminating period of a PWM output, the corresponding pin on the STM32G4 microcontroller should be set HIGH by default, and remain always on.

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